

Networking 1 — Defense & Oral Panel Notes

Comprehensive Defense Guide & Question Assignment Companion

Panelists: The Dean, Kal, and Miss Almie

Spoken Oral: 25 Points Total (5 Qs × 0–5 avg from 3 panelists) breaks without it)

Written Grading: 4 marks each (Decision + Reason + What

Deliverable: PDF list of members & assigned questions submitted

SCORING RUBRIC & PROTOCOL

How Answers Are Scored & Critical Defense Rules

Important Rules:

- Each written answer is worth **4 marks**: requires (1) decision made, (2) reason, and (3) what breaks without it on your own numbers.
- At the oral panel: **~2 minutes per answer out loud with nothing to read from.**
- **Penalty rule:** You will lose points when a designated member does not answer or cannot answer. Another member can rescue/answer, but you will **not get a 5 (maximum 4)** depending on answer quality.
- **Grading note:** Sizing is the written part that Kal grades in person until segmentation; Kal will grade the rest at home.

Score	What the Panel Heard
5	The decision, the reason, and what would break without it, on your own numbers.
4	The decision and the reason, but one part was thin, or you needed one nudge.
3	Correct but thin, or missing the why.
2	Partly right, or right only after the panel led you to it.
1	Read aloud, recited, or rescued by a teammate.
0	No answer, or an answer that was wrong and stayed wrong.

QUESTION 1

Sizing — Why is each subnet the size it is?

Part 1: Write one sentence for each department: how many machines it has, how many addresses it needs once you count the gateway and switch, and which mask gives you that many. (Computer Lab also holds the server = needs one more).

Part 2: If your biggest department doubled, would your plan still fit? Say yes or no. Name the first thing you would change.

PANEL QUESTIONS (YOU WILL ALSO BE ASKED):

- "Walk me through your biggest department. How many machines, and how many addresses did that need?"
- "Why not one size smaller?"
- "What would you have to change if that department doubled?"

TARGET ANSWERS & SCRIPT

Walkthrough of Biggest Department:

The biggest department is the **Admin Office with 29 machines**. It needs **31 usable host addresses** total: 29 for computers, 1 for the Default Gateway (`172.16.6.1`), and 1 for the S-ADMIN Switch (`172.16.6.2`). To accommodate this, it uses a **/26 mask (255.255.255.192)**, providing a total block size of 64 addresses (62 usable).

Why not one size smaller?

A `/28` gives only 14 usable addresses and the Admin Office needs at least 31 (or 27+). Even a `/27` provides only 30 usable addresses, which fails to fit 31 required hosts. Therefore, **/26 is the smallest size that fits**. (*The exact same test decides all departments*).

If that department doubled, what would you have to change?

No, the plan would no longer fit. If doubled (approx. 58 machines → ~60+ host addresses + network/broadcast), it exhausts the /26 subnet. You would have to **re-IP the network / allocate a /25 subnet** because the original block has no remaining room for that level of growth.

QUESTION 2

Segmentation — Campus Subnets & VLANs

Part 1: Name two things your team gains by splitting the campus into three subnets instead of one (practical campus reasons, not generic definitions).

Part 2: Imagine all three departments now share one switch. What would you use to keep them apart? What would a PC in one department need to reach another?

PANEL QUESTIONS (YOU WILL ALSO BE ASKED):

- "If one department has a problem, what stops it reaching the other two?"
- "If all three departments shared one switch, what would you use to keep them apart?"
- "Two of those departments now need to talk to each other. What makes that possible?"

TARGET ANSWERS & SCRIPT

If one department has a problem, what stops it reaching the other two?

The **Router**. A router breaks up broadcast domains; if a broadcast storm, malformed traffic, or network loop occurs in one department, the router drops the broadcast and prevents it from spreading to the other subnets.

If all three departments shared one switch, what would you use to keep them apart?

We would configure separate **VLANs (Virtual Local Area Networks)** for the Admin Office, Faculty Office, and Computer Lab to logically isolate layer-2 broadcast traffic on the shared switch.

Two of those departments now need to talk to each other. What makes that possible?

We install a **Router (Inter-VLAN routing / Router-on-a-Stick)**. Each PC must be configured with its department's **Default Gateway** so packets bound for other subnets/VLANs are forwarded to the router for Layer 3 switching.

QUESTION 3

Security — Device Hardening & Limitations

Part 1: Pick three settings made in Week 9. Write the setting and what somebody could have done if you had not set it.

Part 2: Name one thing a real campus would add that you did not, and why it is out of scope.

PANEL QUESTIONS (YOU WILL ALSO BE ASKED):

- "In Week 9 you typed a set of settings onto every device to protect it. Pick one of those. What could somebody have done if you had not set it?"
- "Which of the settings you typed would a real campus not rely on, and why?"
- "Name one thing a real campus would add that you did not. Why is it out of scope here?"

TARGET ANSWERS & SCRIPT

Week 9 Hardening Settings & Consequences:

- `service password-encryption` : Encrypts plaintext passwords in `running-config` so shoulder-surfers or unauthorized config viewers cannot read them.
- `enable secret <password>` : Secures privileged EXEC mode with a salted MD5/SHA hash preventing unauthorized administrative control.
- `banner motd # Authorized Access Only #` : Provides formal legal warning that unauthorized access is prohibited (essential for legal prosecution).
- `line console 0 / password / exec-timeout 5 0` : Limits local physical console access and terminates inactive sessions after 5 minutes to prevent session hijacking.
- `line vty 0 15 / password / exec-timeout 5 0` : Protects telnet/remote access lines and automatically logs off abandoned remote sessions after 5 minutes.

What would a real campus not rely on, and what would it add instead?

A real campus would not rely on **weak Type 7 password encryption or basic unencrypted Telnet/VTY passwords**. Instead, a real campus would deploy a **Dedicated Firewall, Centralized Authentication (RADIUS/TACACS+), and SSH**. These were out of scope because this is an introductory foundational course (Networking 1), and those advanced enterprise implementations belong to subsequent networking courses.

QUESTION 4

Limits — Fragility & Single Points of Failure

Part 1: Name two things that would take part of your network down if they failed, and what stops working.

Part 2: If SWU added a fourth department to this campus, what would you run out of first?

PANEL QUESTIONS (YOU WILL ALSO BE ASKED):

- "If your one router failed, what would still work and what would not?"
- "Each of your departments depends on one switch. What happens to that department if the switch fails?"
- "If SWU added a fourth department to this campus, what would you run out of first?"

TARGET ANSWERS & SCRIPT

Single Points of Failure & Impact:

1. Single Central Router: If it fails, intra-subnet local communication within a single switch still works, but **the Admin Office and Faculty Office can no longer communicate with each other**, and all three departments lose access to external networks, Internet, and inter-VLAN services.

2. Department Access Switch: If a department switch goes down, **the entire department is completely disconnected**—devices cannot reach each other or the router. Redundant switches/uplinks are needed to eliminate this single point of failure.

If SWU added a fourth department, what would you run out of first?

The **Cisco 2911 Router interfaces / physical ports** because the router does not have enough native routed GigabitEthernet interfaces to dedicate one per department without re-architecting into sub-interfaces (trunking) or adding HWIC expansion modules.

QUESTION 5

Ethics — Privileged Access & Incident Handling

Part 1: Where is the line between having access to a network and being allowed to use it? Name one written rule looked up and how it applies.

Part 2: You find a colleague's password sitting in a config file you were handed. What do you do, and in what order?

PANEL QUESTIONS (YOU WILL ALSO BE ASKED):

- "What rule did you look up, and where did you find it?"
- "How does that rule apply to your own project file?"
- "You find a colleague's password in a file you were handed. Walk me through what you do, in order."

TARGET ANSWERS & SCRIPT

Rule on Access vs. Authorization:

Principle of Least Privilege / Acceptable Use Policy (ACM/IEEE Code of Ethics): Mere technical access does not equal legal or ethical authorization. In our project:

- Keep configuration files strictly within the team; never upload sensitive configs or topology files to open shared drives.
- Never disclose administrative device passwords to anyone outside authorized personnel.
- If credentials or access are ever breached, rotate/change passwords immediately.

Password Discovery Procedure (In Order):

Step 1: Immediately notify that colleague directly so they can change their password and credentials across all systems.

Step 2: Sanitize or securely redact the plaintext credential from the configuration file so it does not propagate.

Step 3: Report the exposure to the supervisor/administrator so the institution can assess risk, audit logs, and implement corrective security safeguards.